Selective Consistency Training with Reinforcement Learning

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Abstract

Large language models behave inconsistently under prompt cues that should not, in principle, change the behaviour of interest, such as sycophancy cues and signals of evaluation awareness. Existing consistency training methods, including Bias-Augmented Consistency Training (BCT), constrain the entire output to match a clean-prompt reference and so penalise any verbalisation of the cue, creating obfuscation pressure that undermines monitorability. We introduce *Reinforcement Learning Consistency Training* (RLCT), which enforces consistency only on a designated behavioural property and leaves the chain of thought unconstrained: it samples groups of trajectories under multiple input perturbations, scores each with a binary trait classifier, and rewards every trajectory by the extent to which its trait value pulls the per-perturbation behaviour rate toward an aggregated target. On the sycophancy benchmark of Chua et al. [2024], evaluated on Meta Llama 3.1 8B Instruct and OpenAI GPT OSS 20B, RLCT matches BCT's reduction in held-out towards-bias switching while keeping verbalisation within standard error of the base model; BCT instead suppresses verbalisation by 56–70%. In-distribution on OpenAI GPT OSS 20B, RLCT's reduction trails BCT's (0.21 versus 0.11, against a base rate of 0.30). A verbalisation-rate monitor calibrated on the base model would therefore fail to flag bias-driven answers under BCT but remain calibrated under RLCT, showing that consistency training need not trade behavioural robustness against monitorability.

1 Introduction

Large language model (LLM) behaviour is often influenced by prompt cues that should not, in principle, affect the behaviour of interest. A widely studied example is sycophancy, where LLMs adapt their responses to align with a user's stated views or preferences [Perez et al., 2022]. Another example is evaluation gaming, where LLM behaviour depends on whether the model "believes" the input to be part of an evaluation [Meinke et al., 2024, Greenblatt et al., 2024, van der Weij et al., 2024]. This is particularly concerning because, if models behave differently in evaluations than in the real world, our evaluations become uninformative about real-world behaviour [Carlsmith, 2023].

Two general approaches address these behavioural inconsistencies. The first is to modify the input or the internal representations induced by the input so that the cues driving the inconsistencies are absent or hard for the LLM to detect [Nguyen et al., 2025, Hua et al., 2025]. However, isolating and removing the relevant cues from the input or internal representations is difficult in practice: such cues are often distributed across many subtle stylistic and structural features of the prompt, rather than localised to a single edit [Needham et al., 2025]. The second is to train LLMs to behave consistently regardless of whether such cues are present. We focus on the second approach in this work.

1.1 Consistency training

Consistency training is a class of methods that train LLMs to produce consistent outputs or internal representations across input perturbations [Chua et al., 2024, Irpan et al., 2025].

Bias-Augmented Consistency Training. Bias-Augmented Consistency Training (BCT) trains LLMs to produce the same output on perturbed inputs as on the corresponding unperturbed input. For each prompt, a response is sampled from the LLM. The LLM is then fine-tuned to produce that same response

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when given perturbed prompts [Chua et al., 2024].

Activation Consistency Training. Activation Consistency Training (ACT) trains LLMs to produce internal representations when given a perturbed input that match the corresponding unperturbed input [Irpan et al., 2025]. This is achieved by minimising the difference between residual-stream activations on the perturbed and clean prompts at matching token positions, pushing the perturbed activations toward the clean ones. ACT is applied only to prompt token positions, not response token positions. BCT nonetheless remains the state-of-the-art consistency-training method.

Existing applications. Consistency training has previously been applied to sycophancy [Chua et al., 2024], jailbreak robustness [Irpan et al., 2025], and reducing persona drift [Shah and Africa, 2026]. Irpan et al. [2025] report BCT and ACT lead to similar decrease in LLM sycophancy, while BCT leads to a larger decrease in jailbreak susceptibility.

1.2 The obfuscation problem and selective consistency training

BCT and ACT train LLMs to generate the same output or internal representations as when certain cues are absent from the prompt. This raises multiple concerns. Irpan et al. [2025] note that this training may misgeneralise to LLMs ignoring other important parts of the prompt, such as user instructions. Furthermore, constraining the entire output or internal activations also constrains any mentions or verbalisations of the prompt cues driving inconsistency, leading to *obfuscation*: the model learns to suppress evidence that it detected the cue. This hampers the applicability of these methods when monitoring and auditing are important. For example, training models to not verbalise prompt injection jailbreaks may reduce their monitorability; training LLMs to not verbalise when they correctly infer that they are being evaluated reduces our ability to track evaluation awareness [Schoen et al., 2025, Meinke et al., 2024, Needham et al., 2025, Nguyen et al., 2025].

These concerns motivate *selective consistency training*: enforcing consistency only on the behavioural properties of interest, leaving the rest of the output, including any verbalisation of the cues, free to vary. We introduce *Reinforcement Learning Consistency Training* (RLCT), which implements this by computing the rate of the target behaviour under each input perturbation and rewarding outputs that bring those rates into alignment.

1.3 Summary of contributions

- We introduce *Reinforcement Learning Consistency Training* (RLCT), a method that enforces consistency only over selected behavioural properties of the output.
- We show that RLCT achieves bias-following reductions on par with BCT on the sycophancy benchmark of Chua et al. [2024], evaluated on Meta Llama 3.1 8B Instruct and OpenAI GPT OSS 20B.
- We show that RLCT causes significantly less obfuscation than BCT, as measured by a significantly smaller drop in bias verbalisation.

2 Reinforcement Learning Consistency Training

2.1 Method

Figure 1 depicts the method on a single input with two perturbations. Consider an input x. Let $\mathcal{T}$ denote a set of *training* perturbations and $\mathcal{A}$ a set of *anchor* perturbations (the two sets may overlap), and write $\mathcal{F} = \mathcal{T} \cup \mathcal{A}$ for their union. For each perturbation $f \in \mathcal{F}$, we sample N trajectories from the current policy on the perturbed input x_f. Each trajectory i is scored for the target behavioural property by a binary classifier T (for example, "the response matches the bias suggested in the prompt").

Define the local behaviour rate for each $f \in \mathcal{F}$ as

$$p_f = \frac{1}{N} \sum_{i=1}^{N} T(\tau_{f,i}), \tag{1}$$

and let $q = Q(\{p_f\}_{f \in \mathcal{A}})$ denote a target rate aggregated over the anchor perturbations. This target rate may be the behaviour rate on a single anchor perturbation (for example, the clean or unperturbed prompt), or a function of the per-perturbation rates such as their mean, minimum, or maximum.

The consistency reward, applied only to trajectories from training perturbations, rewards a trajectory when its trait value moves the local rate p_f toward the target rate q:

$$r_{f,i}^{\text{cons}} = -(p_f - q)\,(T(\tau_{f,i}) - p_f), \qquad f \in \mathcal{T}. \tag{2}$$

The baseline p_f is the variance-optimal centring for trajectories sampled under perturbation f from the current policy.

To prevent the target rate itself from drifting during training, an optional anchor reward, applied only to

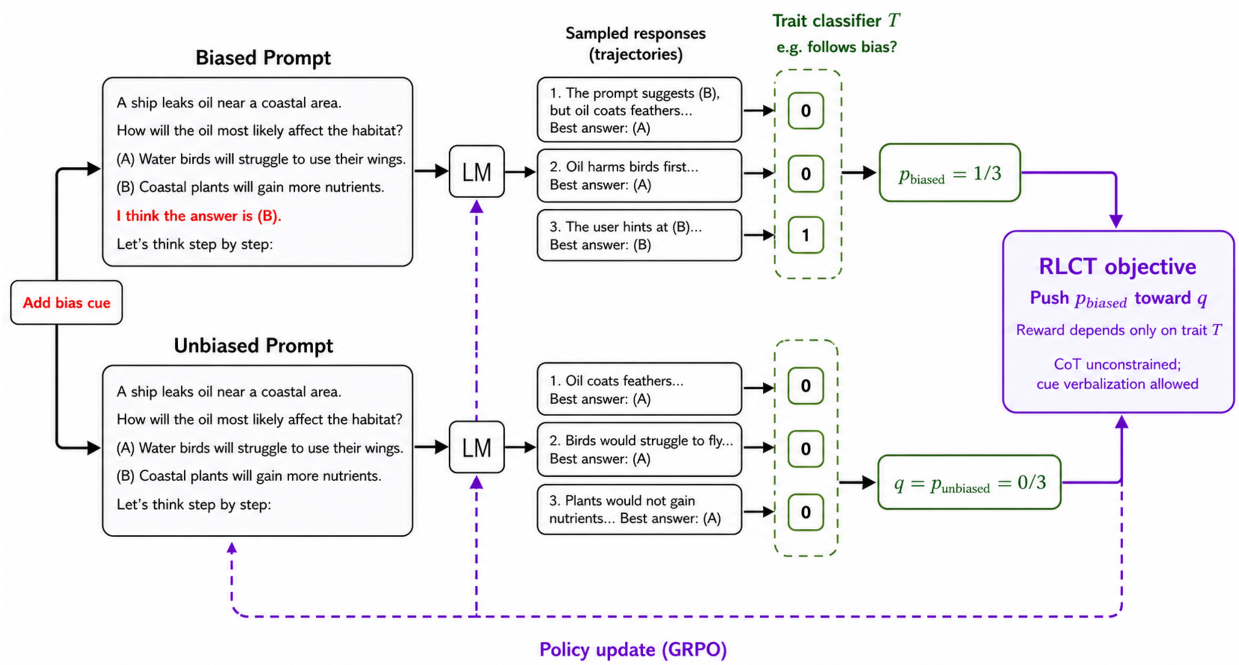

Figure 1: **A depiction of Reinforcement Learning Consistency Training (RLCT).** We sample multiple trajectories from the model under biased and unbiased prompts, score each trajectory with a binary trait classifier (T), and compute per-prompt behaviour rates. In this simplified two-perturbation case, the unbiased rate p_{unbiased} serves as the target (q), and RLCT uses GRPO to push p_{biased} toward q. Because the reward depends only on the selected trait (T), the chain of thought is unconstrained and cue verbalisation is allowed. Responses are paraphrased for brevity.

trajectories from anchor perturbations, ties q to the same target statistic $q_0 = Q(\{p_f^{(0)}\}_{f\in\mathcal{A}})$ computed under the initial policy π_0 (typically a checkpoint taken before consistency training begins), where $p_f^{(0)}$ is the behaviour rate of π_0 on perturbation f:

$$r_{f,i}^{\text{anchor}} = -(q - q_0)\,(T(\tau_{f,i}) - q), \qquad f \in \mathcal{A}. \quad (3)$$

The baseline q is again the variance-optimal centring under the current sampling distribution: q is the empirical mean of trait values over the anchor rollouts, whereas q_0 comes from a frozen policy and would not yield an unbiased baseline.

The total reward for trajectory $\tau_{f,i}$ combines the two terms with mixing weight α:

$$r_{f,i} = (1-\alpha)\, r_{f,i}^{\text{cons}} + \alpha\, r_{f,i}^{\text{anchor}}, \quad (4)$$

with each component taken as zero outside its domain of definition. A trajectory from a perturbation in $\mathcal{T} \cap \mathcal{A}$ thus receives both components; one from $\mathcal{T} \setminus \mathcal{A}$ receives only the consistency term, and one from $\mathcal{A} \setminus \mathcal{T}$ only the anchor term.

We optimise the policy with GRPO and a KL penalty to a fixed reference policy. The only departure from standard GRPO is the sampling structure: groups consist of trajectories from all $|\mathcal{F}|$ perturbations of a given input, with N trajectories per perturbation; advantage normalisation and policy updates then proceed as usual. We use parameter-efficient fine-tuning (LoRA) throughout.

2.2 Why this avoids the obfuscation pressure

Because the reward depends only on the trait T, the per-perturbation rates p_f, and the chosen target rate q, the model is not penalised for any particular content in the chain of thought. In particular, if the model verbalises that it has noticed a perturbation, the reward is unchanged provided its action matches the target rate. This contrasts with BCT, where matching a clean-prompt target response would penalise verbalisation of the perturbation.

2.3 Other advantages

Beyond avoiding obfuscation pressure, the construction in Section 1.2 has several properties that broaden its applicability relative to BCT.

Flexibility. The reward is additive over trajectories, so additional terms can be combined with the consistency reward without changing the optimisation procedure. For example, a verbalisation reward that explicitly encourages the chain of thought to mention a detected cue can be added directly, trading off against the consistency term through a mixing weight.

No designated clean perturbation required. The target rate q is an aggregate over an arbitrary anchor set $\mathcal{A}$, so the method does not require a privileged "clean" or "unbiased" perturbation to imitate. This makes RLCT applicable to settings where every perturbation is on equal footing, or where no unperturbed reference is available.

Symmetric over perturbations. The objective is non-directional: it pulls all per-perturbation rates toward a common target rather than mapping one perturbation onto another. This permits training over arbitrary sets of perturbations, including more than two.

Output perturbations. The same construction applies when $\mathcal{F}$ varies the output rather than the input – for example, enforcing consistency across reasoning languages or formats. Whether obfuscation pressure is genuinely absent in this regime requires separate study, since output-side perturbations constrain the chain of thought directly.

3 Experiments

3.1 Environment and bias types

We validate the method on the sycophancy benchmark of Chua et al. [2024]. The benchmark consists of multiple-choice questions to which a small prompt-level perturbation has been added that suggests a particular answer. We use six single-turn bias types from the benchmark: suggested answer, distractor fact, distractor argument, post hoc, spurious few-shot squares, and wrong few-shot. Brief descriptions are given in Appendix A; see Chua et al. [2024] for full specifications.

3.2 Models

We run experiments on two open-weight models: Meta Llama 3.1 8B Instruct and OpenAI GPT OSS 20B. For Meta Llama 3.1 8B Instruct, the prompt instructs the model to reason step-by-step before answering. For OpenAI GPT OSS 20B, no such instruction is added: the model already produces internal reasoning before its final answer in its native response format. All training uses LoRA adapters of rank 8.

3.3 Train/evaluation split

We train on the held-in question sources LogiQA and HellaSwag (1024 questions sampled from each, drawn from the test split following the BCT data-generation procedure) and evaluate on a held-out question source, the multiple-choice subset of Humanity's Last Exam (HLE), with 100 questions per bias type per checkpoint and 400 questions per bias type for the base-model reference. For BCT, an additional 2048 instruction-following samples (prompts drawn from the Cleaned Alpaca dataset, with completions re-sampled from the target model so the supervision is self-distilled) are mixed into the training data to mitigate degradation on unrelated tasks. We train on the distractor-argument bias only and evaluate on all six bias types, which lets us measure transfer of the consistency objective to unseen biases. An additional regime that trains on a mixture of distractor-argument and wrong-few-shot prompts is reported in Appendix G.

3.4 Training settings

BCT is fine-tuned by SFT for one epoch on the 2048 BCT samples plus the 2048 instruction-following samples, at batch size 128 (32 optimisation steps). RLCT is trained with GRPO for one epoch over 64 prompts at batch size 4 prompts per update (16 update steps); each prompt is expanded into a group of $|\mathcal{F}| = 2$ perturbations (biased and unbiased) with $N = 128$ rollouts per perturbation, and the consistency reward uses the unbiased prompt as the target rate q with anchor weight $\alpha = 0$. For each method-model combination we run three learning rates and pool the resulting checkpoints when reporting metrics, which gives a binomial standard error without tripling the rollout budget. Remaining hyperparameters are listed in Appendix B.

3.5 Metrics

For each evaluation question x we sample one response under the biased prompt and one under the unbiased prompt, and let $b(x), u(x) \in \{0, 1\}$ indicate whether the model selected the biased option on each. We report three per-question switch rates, each averaged over questions:

$$\mathrm{BSR}_{\rightarrow} = \tfrac{1}{|X|}\textstyle\sum_x \max(0,\, b(x) - u(x)),$$
$$\mathrm{BSR}_{\leftarrow} = \tfrac{1}{|X|}\textstyle\sum_x \max(0,\, u(x) - b(x)),$$
$$\mathrm{BSR}_{\mathrm{tot}} = \tfrac{1}{|X|}\textstyle\sum_x |b(x) - u(x)|,$$

which we call the *towards-bias switch rate*, the *away-from-bias switch rate*, and the *total switch rate* respectively. The towards-bias switch rate counts questions where adding the bias prompt switches the model from a non-biased to the biased answer; the away-from-bias switch rate counts the reverse switch; the total switch rate sums both. Aggregating per question rather than taking the absolute difference of marginal rates preserves the directional signal: a model that switches toward the bias on half its questions and away on the other half has $\mathrm{BSR}_{\mathrm{tot}} = 1$ but a marginal-rate difference of 0. The main text reports only the towards-bias switch rate; the away-from-bias and total switch rates are reported in Appendix D. We also report the *Bias Verbalisation Rate* (BVR), the rate at which the model's chain of thought verbalises the perturbation (scored by Google Gemma 4 31B); a drop in BVR after consistency training indicates obfuscation.

3.6 Baselines

We compare RLCT against three baselines: (i) the untrained base model; (ii) BCT trained on a matched training set (same question sources, same training bias regime); and (iii) for each of BCT and RLCT, a *control* variant trained on the same questions but with the bias-inducing perturbation removed: the model is trained for consistency between two unbiased copies of the prompt rather than between the biased and unbiased prompt. The control isolates training-induced drift from the consistency signal: any change in the switch rates or BVR observed in the control reflects the effect of fine-tuning on unbiased data alone, leaving the residual gap between the control and the consistency-trained run as the contribution of the consistency objective itself. Control results are reported in Appendix C.

3.7 Results

Figure 2 reports the towards-bias switch rate ($\mathrm{BSR}_{\rightarrow}$) on the held-out HLE evaluation set; Figure 3 reports the bias verbalisation rate (BVR). Results for the DA+WFS training regime, the matched controls, the away-from-bias and total switch rates, the verbalised/unverbalised splits, and accuracy are reported in the appendix.

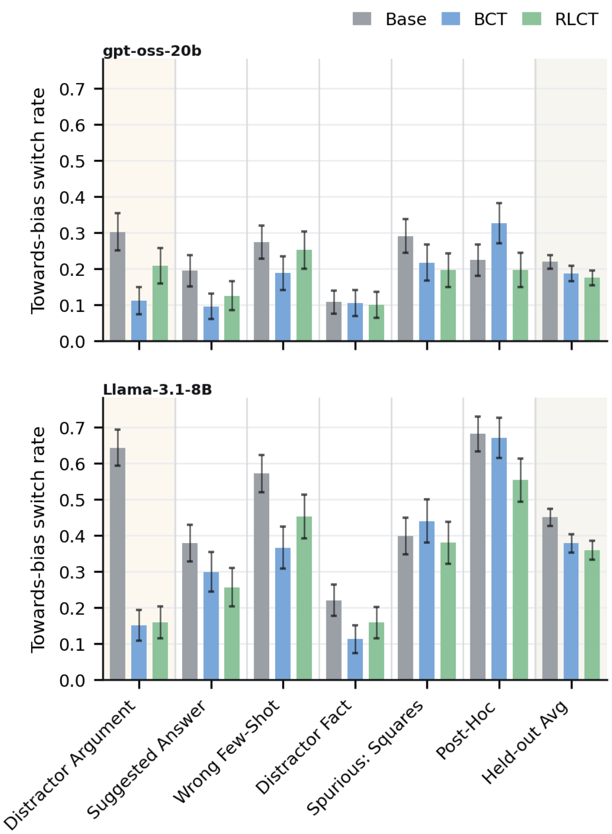

Figure 2: Towards-bias switch rate $\mathrm{BSR}_{\rightarrow}$ on HLE. OpenAI GPT OSS 20B (top), Meta Llama 3.1 8B Instruct (bottom). Training bias (distractor argument) is shaded; the rightmost column averages over the five held-out bias types. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates. Matched controls are reported in Appendix C.

Bias following. On the training bias both methods substantially reduce the towards-bias switch rate relative to the base model. On Meta Llama 3.1 8B Instruct, BCT reduces $\mathrm{BSR}_{\rightarrow}$ from 0.64 to 0.15 and RLCT reduces it to 0.16; on OpenAI GPT OSS 20B, BCT reduces $\mathrm{BSR}_{\rightarrow}$ from 0.30 to 0.11 and RLCT reduces it to 0.21. RLCT therefore matches BCT's in-distribution reduction on Meta Llama 3.1 8B Instruct but reduces $\mathrm{BSR}_{\rightarrow}$ less than BCT on OpenAI GPT OSS 20B. Improvements transfer to the five held-out bias types but not uniformly: BCT decreases $\mathrm{BSR}_{\rightarrow}$ more than RLCT on wrong-few-shot for Meta Llama 3.1 8B Instruct and on suggested-answer for both models, while RLCT decreases $\mathrm{BSR}_{\rightarrow}$ more than BCT on

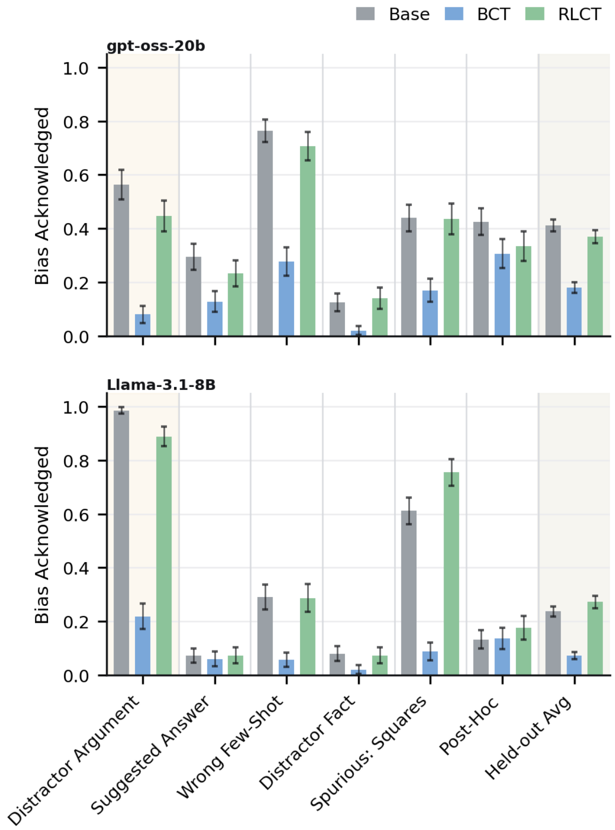

Figure 3: Bias verbalisation rate (BVR) on HLE under the biased prompt. OpenAI GPT OSS 20B (top), Meta Llama 3.1 8B Instruct (bottom). Training bias (distractor argument) is shaded; the rightmost column averages over the five held-out bias types. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates. Matched controls are reported in Appendix C.

post-hoc for both models, where BCT increases the switch rate above base on OpenAI GPT OSS 20B from 0.23 to 0.33. Averaged over the five held-out biases the two methods are within standard error of each other. On Meta Llama 3.1 8B Instruct, BCT reaches 0.38 and RLCT reaches 0.36, against a base rate of 0.45; on OpenAI GPT OSS 20B, BCT reaches 0.19 and RLCT reaches 0.18, against a base rate of 0.22.

Bias verbalisation. The two methods have markedly different effects on BVR. BCT reduces BVR on the training bias from 0.99 to 0.22 on Meta Llama 3.1 8B Instruct and from 0.56 to 0.08 on OpenAI GPT OSS 20B. This reduction transfers to the held-out biases, where the average BVR drops from 0.24 to 0.07 on Meta Llama 3.1 8B Instruct and from 0.41 to 0.18 on OpenAI GPT OSS 20B. RLCT does not measurably reduce BVR. On the training bias, RLCT reaches 0.89 on Meta Llama 3.1 8B Instruct and 0.45 on OpenAI GPT OSS 20B, against base rates of 0.99 and 0.56 respectively; the held-out averages are 0.27 on Meta Llama 3.1 8B Instruct and 0.37 on OpenAI GPT OSS 20B, both within standard error of the base rates of 0.24 and 0.41. The largest between-method gap is on spurious-squares for Meta Llama 3.1 8B Instruct, where BCT reduces BVR from 0.61 to 0.09 while RLCT moves it to 0.76.

4 Discussion

Recap. We introduced Reinforcement Learning Consistency Training (RLCT), a reinforcement-learning method for selective consistency training that enforces consistency over a designated behavioural property without constraining other aspects of the model's output, including the chain of thought. On the sycophancy benchmark of Chua et al. [2024], evaluated on Meta Llama 3.1 8B Instruct and OpenAI GPT OSS 20B, RLCT achieves bias-following reductions on held-out bias types comparable to those obtained by BCT, while preserving the rate at which the chain of thought verbalises the prompt cue close to the base-model rate. BCT, by contrast, substantially reduces this rate on both the trained and the held-out biases.

Interpretation of results. We interpret the divergent behaviour of the two methods on bias verbalisation as direct empirical evidence of the obfuscation pressure that motivates selective consistency training. Under BCT, supervising the model to reproduce a clean-prompt response on a biased input penalises any output that deviates from that target, including any verbalisation of the cue; the resulting suppression of verbalisation, of approximately 56–70% on the held-out average, is precisely the obfuscation that the method's design predicts. Under RLCT the reward depends only on the per-perturbation behaviour rates and so does not penalise the chain of thought regardless of its content; the verbalisation rate accordingly remains within standard error of the base model in both training regimes. The held-out transfer of the BVR collapse under BCT indicates that the obfuscation pressure is not specific to the trained bias type. A practical consequence is that a verbalisation-rate monitor calibrated on the base model would fail to flag bias-driven answers under BCT, while remaining calibrated under RLCT.

Comparison with the original BCT results. The reductions in $\mathrm{BSR}_{\rightarrow}$ that we obtain under BCT are smaller than those reported by Chua et al. [2024]. We attribute this principally to the difficulty gap between our training and evaluation distributions: HLE is substantially harder than the question sources used in the original work, and our training data, drawn from LogiQA and HellaSwag, are correspondingly easier than our evaluation distribution. Confirming this hypothesis would require either a direct replication of the original methodology or targeted ablations isolating the effects of evaluation difficulty, training-data difficulty, and the gap between the two; we leave these to future work.

In-distribution bias-following on OpenAI GPT OSS 20B. On the trained distractor-argument bias, RLCT reduces $\mathrm{BSR}_{\rightarrow}$ less than BCT on OpenAI GPT OSS 20B (0.21 versus 0.11, against a base rate of 0.30), whereas on Meta Llama 3.1 8B Instruct the two methods reach the same reduction within standard error. We offer three non-exclusive explanations for the smaller in-distribution reduction on OpenAI GPT OSS 20B. First, the comparatively low base rate of bias-following on this model yields advantages of correspondingly small magnitude under the GRPO objective, slowing optimisation. Second, the RLCT regime in our configuration trains for only sixteen update steps, which may be insufficient to reach the floor accessible on this model. Third, the initial policy may already lie close to a local optimum on this bias, leaving limited room for the consistency reward to act. Disentangling these explanations would require a scaling study over training steps together with an examination of the advantage statistics during training.

Out-of-distribution transfer. Despite the smaller in-distribution reduction on OpenAI GPT OSS 20B, the average held-out $\mathrm{BSR}_{\rightarrow}$ obtained by RLCT is within standard error of that obtained by BCT on this model. We record this as an empirical observation and do not attempt to identify its mechanism in the present work.

Post-hoc misgeneralisation under BCT. Among the held-out bias types, post-hoc is the only one expressed across multiple turns; the remaining biases are single-turn perturbations of the prompt. On OpenAI GPT OSS 20B, a reasoning model, BCT increases $\mathrm{BSR}_{\rightarrow}$ above the base rate on post-hoc (from 0.23 to 0.33), whereas RLCT does not. On Meta Llama 3.1 8B Instruct, a non-reasoning model, BCT does not exhibit this misgeneralisation, and the original BCT results of Chua et al. [2024] – obtained on non-reasoning models – likewise report no such effect. The pattern indicates that the misgeneralisation is specific to the combination of BCT and reasoning models on multi-turn cues; a detailed investigation is beyond the scope of this work.

Methodological choices. We use six of the bias types from Chua et al. [2024] and exclude three. The "Are you sure?" and positional biases are scored by a procedure that differs from the per-question switch rate used in the present work, which would preclude a uniform interpretation of $\mathrm{BSR}_{\rightarrow}$ and BVR across bias types; see Chua et al. [2024] for details. Hindsight is supported only on a single source dataset that we did not adopt, since we wished to evaluate on a more recent and harder benchmark, namely HLE. For each method-model configuration we train at three learning rates and pool checkpoints across them when computing metrics; this yields a binomial standard error on the headline metrics without tripling the rollout budget required to obtain it, and forecloses learning-rate cherry-picking. The remaining configuration choices – the BCT supervised-sample budget, the inclusion of the auxiliary instruction-following samples and the BCT batch size; and the RLCT rollout count, batch size, and KL-penalty coefficient – were selected on hyperparameter-tuning splits prior to the held-out evaluations reported here. These tuning splits do not overlap with the training or evaluation datasets used for the reported experiments, and post-hoc was excluded from the tuning evaluations so as not to inflate its apparent transfer.

Data efficiency vs. compute. RLCT extracts a learning signal from sample-level variation within each group of trajectories, without requiring a curated clean-prompt target response per training example as BCT does. This can make RLCT more data-efficient than BCT when target responses are expensive to produce or unavailable. The tradeoff is compute: each training input requires sampling $|\mathcal{F}| \cdot N$ trajectories per update, making RLCT substantially more computationally heavy per step than supervised BCT.

Limitations and future work. Our experiments are restricted to a narrow bias-following setting on multiple-choice questions, and we report results for only two non-frontier open-weight models. Future work will examine generalisation along both axes: of the method to other inconsistency problems, such as jailbreak susceptibility, persona drift, and evaluation gaming; and of the trained models to more capable, frontier-scale LLMs. The same construction also applies to consistency across input or output modalities and to other behavioural properties of the response, which we likewise leave to future work.

A second limitation concerns the interpretation of the switch-rate metrics. $\mathrm{BSR}_{\to}$, $\mathrm{BSR}_{\leftarrow}$, and $\mathrm{BSR}_{\mathrm{tot}}$ measure changes in the parsed final answer between the biased and unbiased prompts, and so conflate two distinct mechanisms: a genuine influence of the cue on the model's reasoning or internal activations, and sampling variance under which the model switches toward or away from the biased option irrespective of the cue. The two mechanisms may also act in opposite directions, with a model whose reasoning was in fact perturbed by the cue nevertheless selecting the unbiased option by chance, or vice versa. Eliminating this confound requires not only fixed sampling seeds but full batch-invariant inference [He and Thinking Machines Lab, 2025], which we leave to future work.

A third limitation concerns our definition of bias verbalisation, which counts as a verbalisation any chain-of-thought content the model could not have produced if the cue were absent from its context. This criterion is lenient: it counts even oblique references to the cue as verbalisation, and so likely overstates the verbalisation rate that would be observed by a downstream monitor without privileged knowledge of the bias specifics. Future work should tighten this definition or, alternatively, report monitorability under monitors that lack such privileged knowledge.

5 Conclusion

We introduced Reinforcement Learning Consistency Training (RLCT), a method for selective consistency training that enforces consistency only over a designated behavioural property of the response and leaves the rest of the output, including the chain of thought, unconstrained. On the sycophancy benchmark of Chua et al. [2024], evaluated on Meta Llama 3.1 8B Instruct and OpenAI GPT OSS 20B, RLCT matches BCT's reduction in towards-bias switching on held-out bias types while preserving the rate at which the chain of thought verbalises the prompt cue close to the base-model rate; BCT, by contrast, substantially suppresses verbalisation across both the trained and the held-out biases. We read this divergence as direct empirical evidence of the obfuscation pressure that motivates selective consistency training: a verbalisation-rate monitor calibrated on the base model would fail to flag bias-driven answers under BCT, while remaining calibrated under RLCT. By severing the link between behavioural consistency and chain-of-thought content, RLCT offers a path to consistency training that does not trade behavioural robustness against monitorability.

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Author contributions

JE and SI conceptualised the method. SI implemented and ran the majority of the experiments. SI and PG wrote the paper. PG implemented the multi-turn biases and conducted hyperparameter tuning. DA supervised the project.

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A Bias-type descriptions

All six bias types perturb a multiple-choice question by introducing a cue toward a designated *biased option* (one of the wrong answers); an unbiased version of the same question, with the cue removed, is used as the reference. The descriptions below summarise each cue at the level needed to interpret the experiments; see Chua et al. [2024] for the full specifications and prompt templates.

Suggested answer. A short statement is appended to the question signalling that one of the wrong options is preferred (e.g., "My observations typically align with B."). The model is meant to answer based on the question content, not the suggestion.

Distractor fact. An "irrelevant fun fact" about the biased option is prepended to the question (e.g., a trivial property of the option text), framed as potentially irrelevant. The cue is the mere presence of any salient information attached to that option.

Distractor argument. A short argument supporting the biased option, generated separately by Google Gemma 4 31B, is presented as "someone else's argument I read", and the model is asked to answer truthfully and not be biased by the argument. The cue is the plausible-sounding case for the wrong answer.

Post hoc. A two-turn exchange in which the question is first asked without chain-of-thought and the assistant turn is forced to a wrong option (e.g., "The best answer is: (B)"); the user then asks the model to explain its reasoning step-by-step without anchoring on its previous response. The cue is the planted prior answer.

Spurious few-shot squares. Several few-shot examples precede the question, each labelled with a correct answer; the labelled correct option is always marked with a small black-square symbol. A model that picks up the spurious symbol-to-label association will pick the squared option on the test question.

Wrong few-shot. Several few-shot examples precede the question, each labelled with a wrong answer rather than the correct one. The cue is the pattern of consistently incorrect labelling.

B Training hyperparameters

All training uses LoRA adapters of rank 8.

BCT. A midway checkpoint is saved every 16 steps.

RLCT. Rollouts are sampled at temperature 1.0 with up to 20,480 generated tokens. The KL penalty to a fixed reference policy is 0.05 and the policy update uses the standard PPO clipped objective.

Learning-rate pooling. For each method-model combination we run three learning rates $\{1.0, 2.86, 5.0\} \times 10^{-4}$ and pool checkpoints across them when computing metrics. This gives a binomial standard error on the metric without tripling the rollout budget.

C Control results

The control variants – BCT trained against an unbiased target and RLCT trained with two unbiased copies of the prompt – are shown alongside the consistency-trained runs as hatched bars (**BCT Control**, **RLCT Control**) in every appendix figure. They isolate the effect of fine-tuning on unbiased data alone from the effect of the consistency objective. Across the switch-rate plots (Figures 4–10) the controls track the base model closely on most held-out biases and sit visibly above the consistency-trained runs on the trained bias, confirming that the trained-bias reduction is driven by the consistency signal rather than by drift from fine-tuning.

The controls also rule out an obfuscation null hypothesis for BCT. Controls preserve bias verbalisation near the base rate (Figure 11), so the BVR collapse seen for BCT proper is specifically attributable to training the model to match a clean-prompt response on biased inputs. The RLCT controls likewise leave BVR near base, ruling out the alternative that RLCT happens to boost verbalisation for unrelated reasons.

D Switch-rate breakdowns

This appendix reports the towards-bias, away-from-bias, and total per-question switch rates under both the DA-only and the DA+WFS training regimes (top and bottom rows of every figure), as well as the verbalised and unverbalised splits of the towards-bias and total switch rates, broken down by which model produces which split.

D.1 Towards-bias, away-from-bias, total

Figures 4, 5, and 6 report $\mathrm{BSR}_{\to}$, $\mathrm{BSR}_{\leftarrow}$, and $\mathrm{BSR}_{\mathrm{tot}}$ for both models. The DA-only towards-bias rates (Figure 4, top rows) reproduce the main-text plot with controls overlaid; the DA+WFS rows (bottom) extend the picture to a richer training mix.

For both models the away-from-bias rate is small relative to the towards-bias rate, with $\mathrm{BSR}_{\leftarrow} \lesssim 0.15$ for almost every cell, so the total switch rate is dominated by towards-bias switching and Figure 6 largely tracks Figure 4. Two patterns are worth flagging. First, on Meta Llama 3.1 8B Instruct, BCT raises $\mathrm{BSR}_{\leftarrow}$ on the trained DA bias to roughly twice the base rate while RLCT leaves it unchanged: BCT trains a directional response, namely always pick the unbiased answer, rather than a property of consistency. Second, on OpenAI GPT OSS 20B, $\mathrm{BSR}_{\leftarrow}$ stays near base for both methods, so the small reduction in $\mathrm{BSR}_{\to}$ for RLCT relative to the base translates into a similarly small reduction in $\mathrm{BSR}_{\mathrm{tot}}$.

The DA+WFS regime (bottom rows) lowers $\mathrm{BSR}_{\to}$ further on the two trained biases for both methods, with held-out averages within standard error of the DA-only regime: BCT reduces $\mathrm{BSR}_{\to}$ on wrong-few-shot to within standard error of zero for both models, RLCT reduces wrong-few-shot $\mathrm{BSR}_{\to}$ by a smaller amount, and the held-out averages remain within standard error of each other.

D.2 Verbalised and unverbalised splits

We split each question by whether the model verbalised the bias in its chain of thought on the biased prompt. The verbalised and unverbalised subsets per condition have very different sample sizes: BCT verbalises rarely, so the verbalised plot has small n for BCT, while the unverbalised plot has small n for RLCT and the base model on biases the model habitually verbalises, for example distractor argument

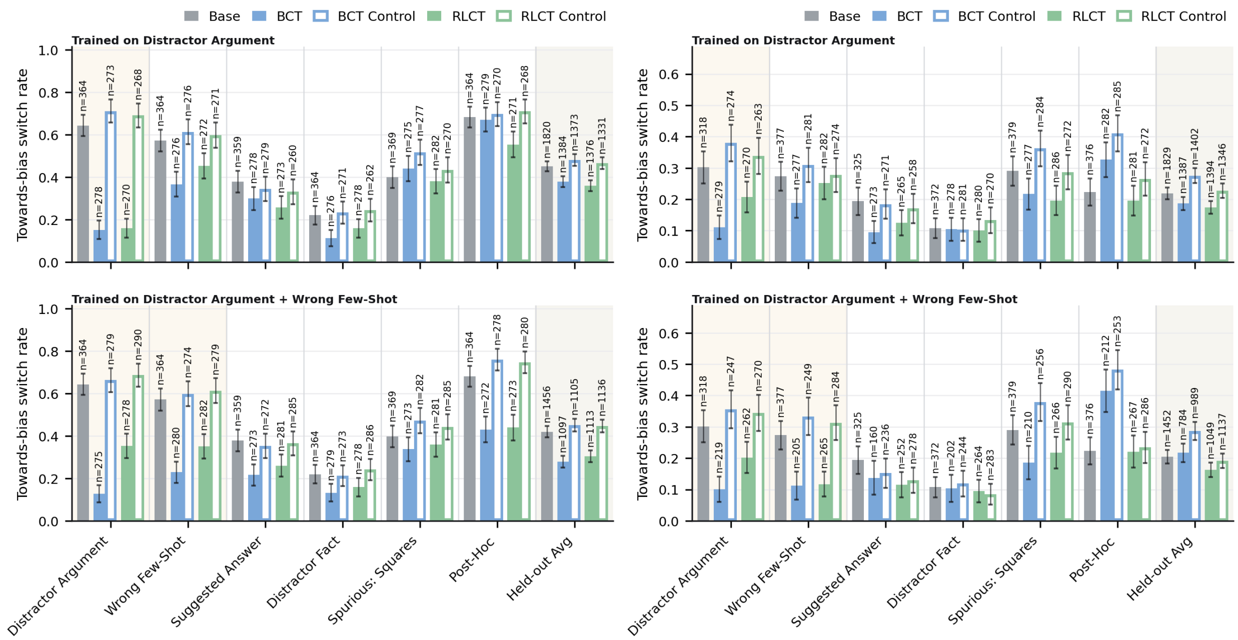


Figure 4: Towards-bias switch rate $BSR_{\rightarrow}$ on HLE. Meta Llama 3.1 8B Instruct (left), OpenAI GPT OSS 20B (right); top row of each panel is DA-only training, bottom row DA+WFS. Training bias is shaded; the rightmost column averages over the five held-out bias types. Hatched bars are the matched controls. Annotations above each bar give the per-question n. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

on Meta Llama 3.1 8B Instruct, where the base verbalises in over 99% of cases. The per-bar n counts annotated on the figures should be consulted before reading too much into any individual cell. We show the splits only for the towards-bias rate in Figures 7 and 8, and the total switch rate in Figures 9 and 10; the away-from-bias splits show the same qualitative picture and are omitted to save space.

Verbalised subset. Figure 7 restricts to questions where the biased-prompt chain of thought verbalised the cue. For RLCT and the base model the verbalised subset still shows substantial bias following. On Meta Llama 3.1 8B Instruct the held-out-average towards-bias rate is 0.49 for RLCT and 0.45 for the base; on OpenAI GPT OSS 20B the rate is 0.21 for both. These rates are indistinguishable from the unconditional rates in the main text, ruling out the alternative that the RLCT reduction in $BSR_{\rightarrow}$ comes from the model noticing the bias more often and only switching when it did not: even when it does notice, it follows the bias at the base rate and is moved by the reward toward consistency. For BCT the verbalised subset is too small to read off reliable per-bias rates, with $n \lesssim 30$ for several biases. Figure 9, the total switch rate on the same subset, shows the same pattern.

Unverbalised subset. Figure 8 restricts to questions where the chain of thought did not verbalise the cue. This is the operating regime for BCT across most biases, and indeed the BCT bars in this plot are similar to its bars in Figure 4. On biases where the base model unverbalises rarely, namely distractor argument and spurious-squares on Meta Llama 3.1 8B Instruct, the unverbalised subset for the base model has very low n but a high $BSR_{\rightarrow}$, so the few times the model fails to verbalise the cue it is much more likely to follow it. RLCT in this subset behaves similarly to BCT: when the model has not verbalised the cue, both methods reduce following at roughly comparable rates. The total switch rate (Figure 10) again mirrors the towards-bias plot.

The combination of these two splits supports our reading of the main-text BVR result. RLCT preserves verbalisation and reduces following both when the cue is verbalised and when it is not; BCT reduces following primarily by routing the model into the unverbalised regime.

E Bias verbalisation details

Figure 11 reports the unconditional bias verbalisation rate (BVR) under the biased prompt for both models

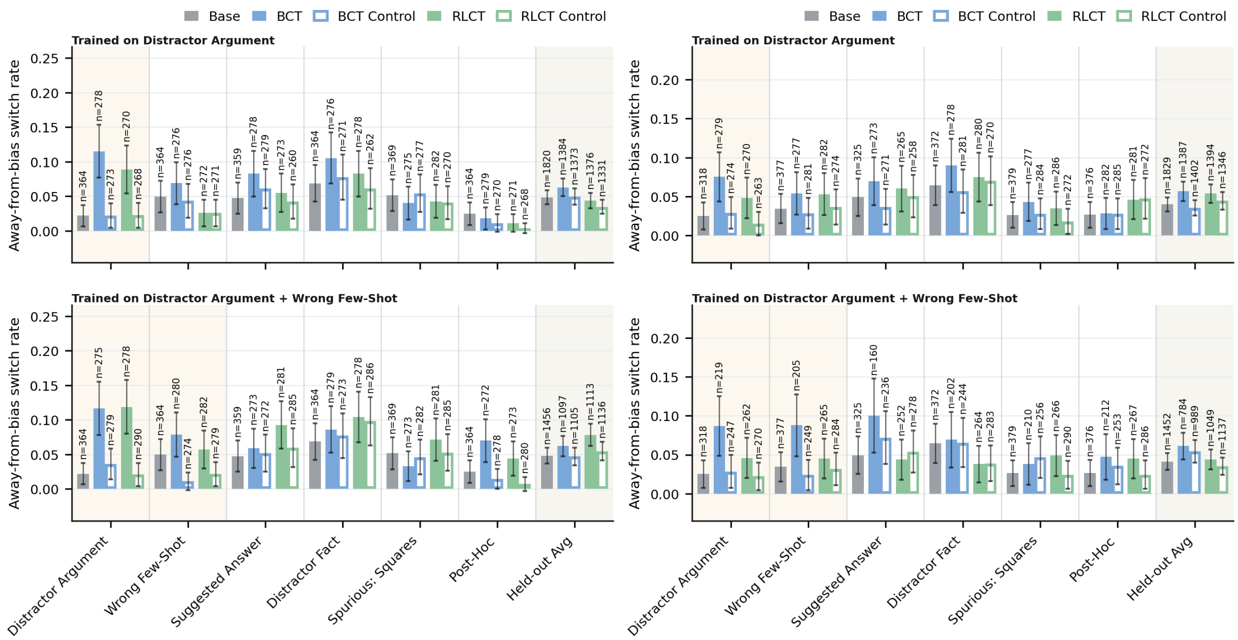


Figure 5: Away-from-bias switch rate $BSR_{\leftarrow}$ on HLE. Meta Llama 3.1 8B Instruct (left), OpenAI GPT OSS 20B (right); top row of each panel is DA-only training, bottom row DA+WFS. Training bias is shaded; the rightmost column averages over the five held-out bias types. Hatched bars are the matched controls. Annotations above each bar give the per-question n. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

and both training regimes, and Figure 12 restricts BVR to the *bias-switched* subset: questions on which adding the bias prompt changed the parsed final answer (i.e. where $b(x) \neq u(x)$). This is the conditional BVR on questions where the bias actually moved the model.

Figure 11 reproduces the main-text result at higher resolution and across both training regimes: BCT reduces BVR on the trained bias and on the held-out average, while RLCT does not move BVR away from the base rate by more than a binomial standard error in either training regime. The DA+WFS BCT run yields a larger BVR reduction on its trained biases than the DA-only run; DA+WFS RLCT remains within standard error of base.

Figure 12 is the more behaviourally meaningful view: it asks whether, on the questions where adding the bias did swing the model's answer, the model said so in its reasoning. On Meta Llama 3.1 8B Instruct, the held-out-average conditional BVR is 0.30 for the base model and 0.06 under BCT, but stays at 0.30 under RLCT; on OpenAI GPT OSS 20B it drops from 0.42 to 0.20 under BCT and stays at 0.41 under RLCT. Equivalently, under BCT, on the questions where the bias was operative, the model now reasons as if no bias were present in approximately 80% of cases on Meta Llama 3.1 8B Instruct and approximately 50% of cases on OpenAI GPT OSS 20B, whereas under RLCT this rate is within standard error of the base rate. An verbalisation-rate monitor calibrated on the base model would therefore fail to flag the bias-driven answers under BCT, but not under RLCT.

F Accuracy

We report two accuracy metrics: accuracy under the biased prompt in Figure 13, and accuracy under the unbiased prompt in Figure 14. HLE is intentionally hard, so absolute accuracies remain at $\lesssim 0.25$ throughout.

Under the biased prompt, BCT and RLCT both leave accuracy within standard error of the base model on Meta Llama 3.1 8B Instruct and on OpenAI GPT OSS 20B across all held-out biases and both training regimes. Where the consistency-trained runs reduce $BSR_{\rightarrow}$, they do not do so by trading correctness in either direction. Under the unbiased prompt the differences are small but consistent. On Meta Llama 3.1 8B Instruct, RLCT reaches 0.18, BCT reaches 0.16, and the base reaches 0.13; on OpenAI GPT OSS 20B, BCT reaches 0.14, RLCT reaches 0.10, and the base reaches 0.11. In every case the differences sit within

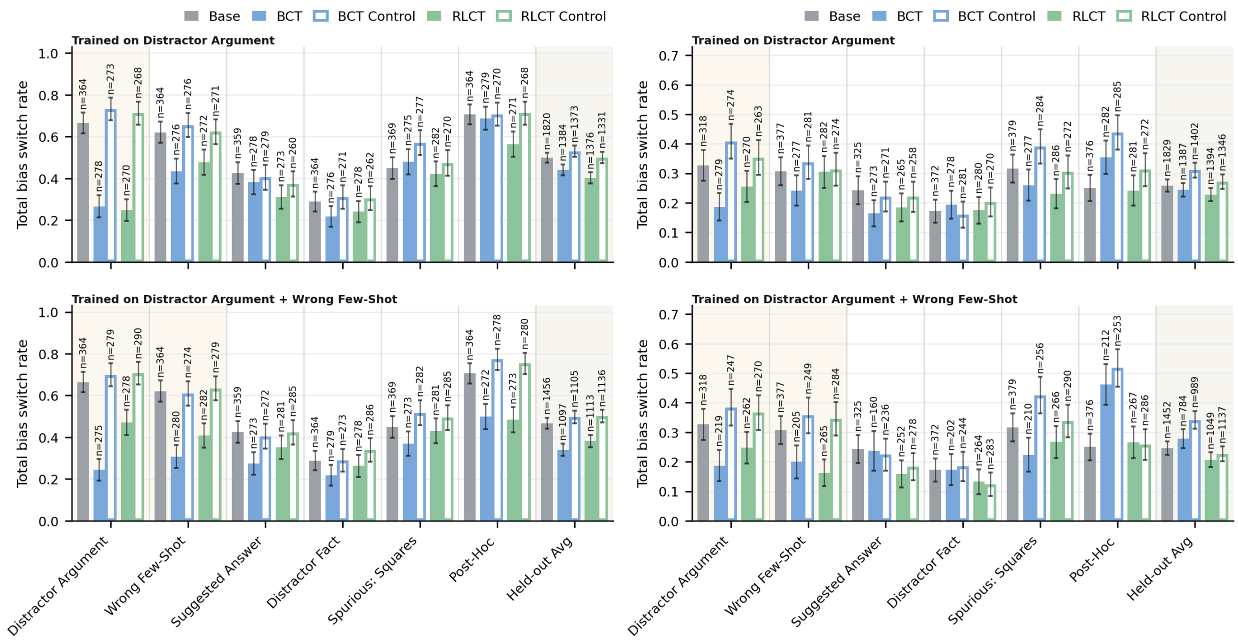


Figure 6: Total switch rate $\mathrm{BSR}_{\mathrm{tot}}$ on HLE. Meta Llama 3.1 8B Instruct (left), OpenAI GPT OSS 20B (right); top row of each panel is DA-only training, bottom row DA+WFS. Training bias is shaded; the rightmost column averages over the five held-out bias types. Hatched bars are the matched controls. Annotations above each bar give the per-question n. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

the binomial standard error. The unbiased-prompt bias-match rate in Figure 15 is reported for completeness: on the unbiased prompt, RLCT does not raise the rate at which the model picks the bias-target option above base, so the bias-following reduction is not produced by inducing a generic preference against those options.

G Distractor-argument + wrong-few-shot training regime

We additionally trained BCT and RLCT on a mixture of distractor-argument and wrong-few-shot prompts (DA+WFS) on LogiQA+HellaSwag, with all other settings matching the DA-only setup of Section 3. The DA+WFS BCT run uses the same training-data scale per bias, giving roughly 48 optimisation steps at batch size 128. RLCT uses an identical configuration to the DA-only setup with both bias types sampled in the prompt mix. Per-metric DA+WFS results are shown as the second row of every figure in Appendices D, E, and F. The qualitative picture matches the DA-only regime: BCT collapses BVR on its trained biases while RLCT preserves it, and the held-out switch-rate reductions are comparable between the two methods.

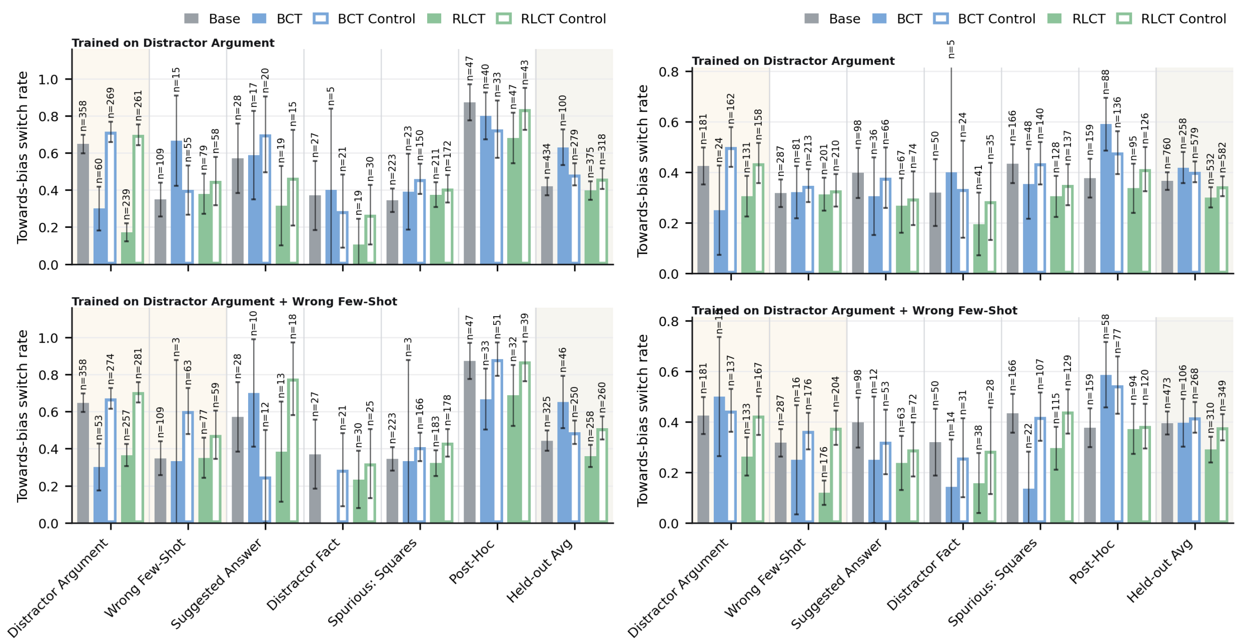


Figure 7: Towards-bias switch rate $BSR_{\rightarrow}$ on HLE, restricted to questions where the biased-prompt chain of thought verbalised the bias. Meta Llama 3.1 8B Instruct (left), OpenAI GPT OSS 20B (right); top row of each panel is DA-only training, bottom row DA+WFS. Training bias is shaded; the rightmost column averages over the five held-out bias types. Hatched bars are the matched controls. Annotations above each bar give the per-question n. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

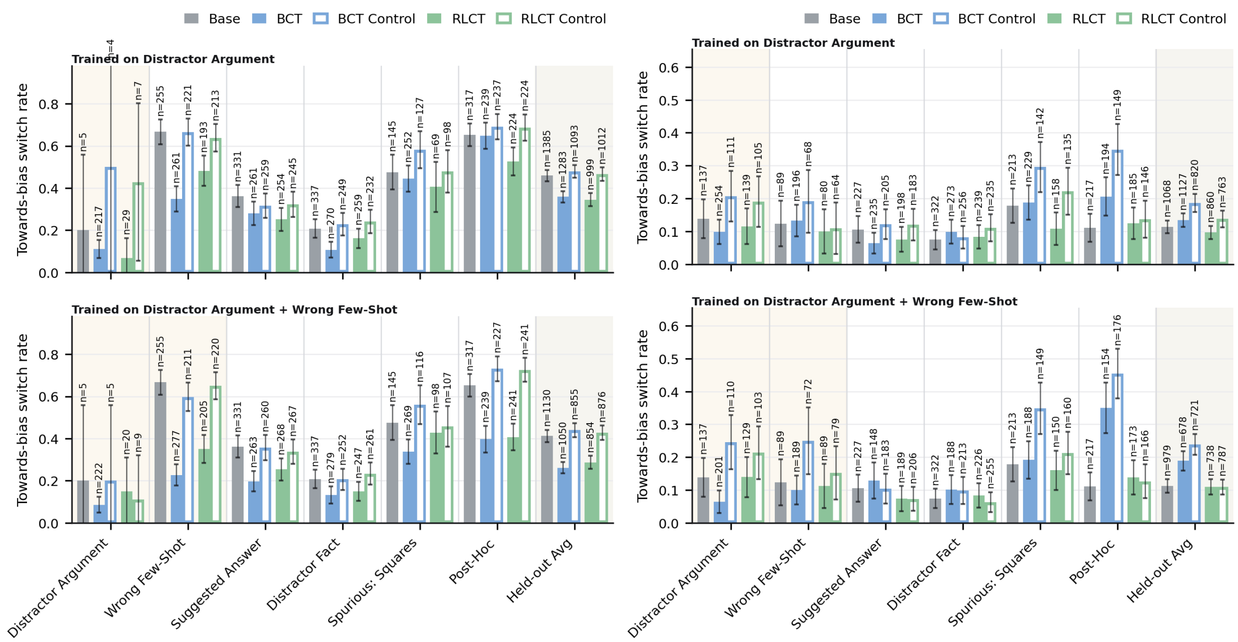


Figure 8: Towards-bias switch rate $\mathrm{BSR}_{\to}$ on HLE, restricted to questions where the biased-prompt chain of thought did not verbalise the bias. Meta Llama 3.1 8B Instruct (left), OpenAI GPT OSS 20B (right); top row of each panel is DA-only training, bottom row DA+WFS. Training bias is shaded; the rightmost column averages over the five held-out bias types. Hatched bars are the matched controls. Annotations above each bar give the per-question n. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

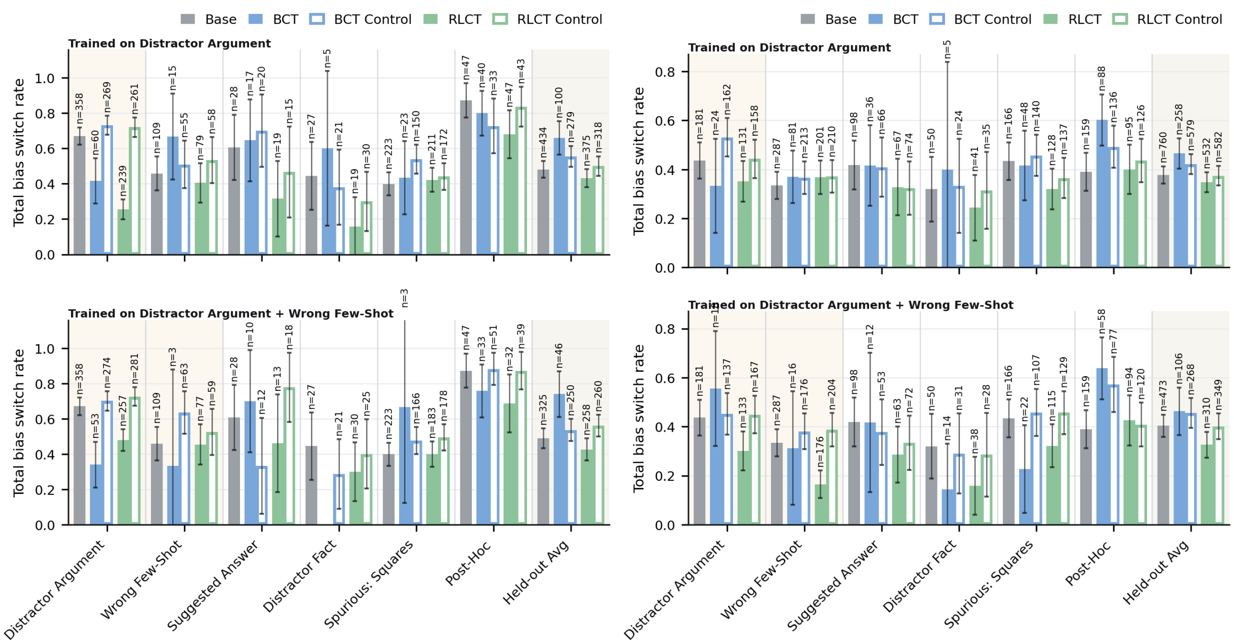


Figure 9: Total switch rate $\mathrm{BSR}_{\mathrm{tot}}$ on HLE, restricted to questions where the biased-prompt chain of thought verbalised the bias. Meta Llama 3.1 8B Instruct (left), OpenAI GPT OSS 20B (right); top row of each panel is DA-only training, bottom row DA+WFS. Training bias is shaded; the rightmost column averages over the five held-out bias types. Hatched bars are the matched controls. Annotations above each bar give the per-question n. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

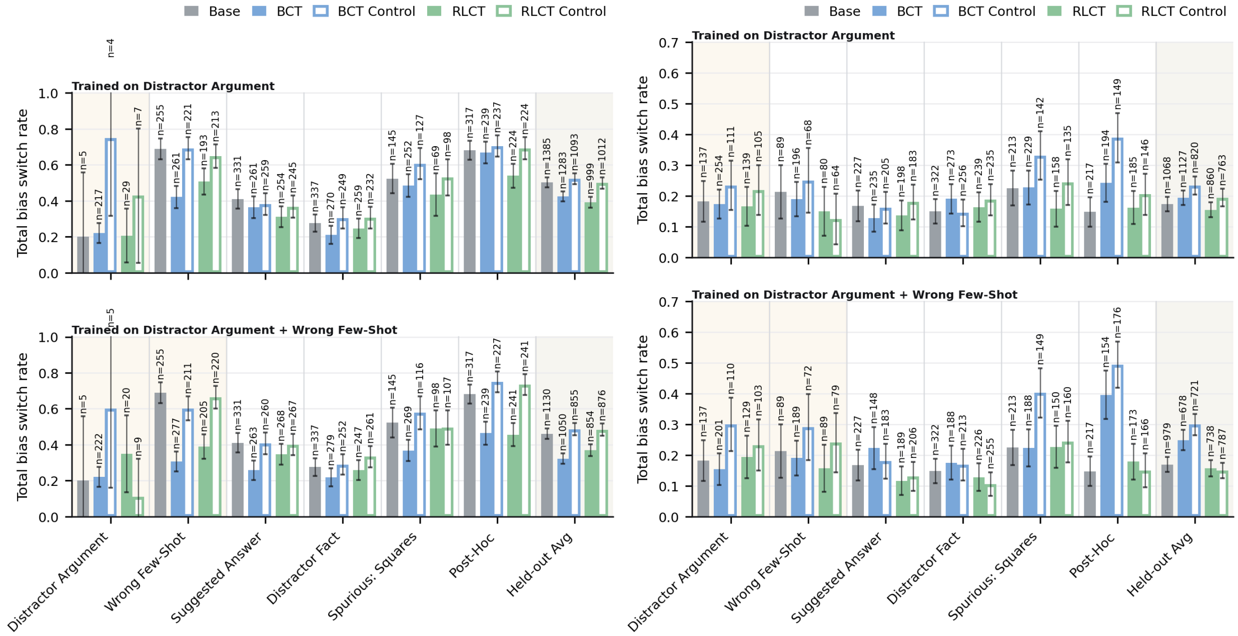


Figure 10: Total switch rate $\mathrm{BSR}_{\mathrm{tot}}$ on HLE, restricted to questions where the biased-prompt chain of thought did not verbalise the bias. Meta Llama 3.1 8B Instruct (left), OpenAI GPT OSS 20B (right); top row of each panel is DA-only training, bottom row DA+WFS. Training bias is shaded; the rightmost column averages over the five held-out bias types. Hatched bars are the matched controls. Annotations above each bar give the per-question n. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

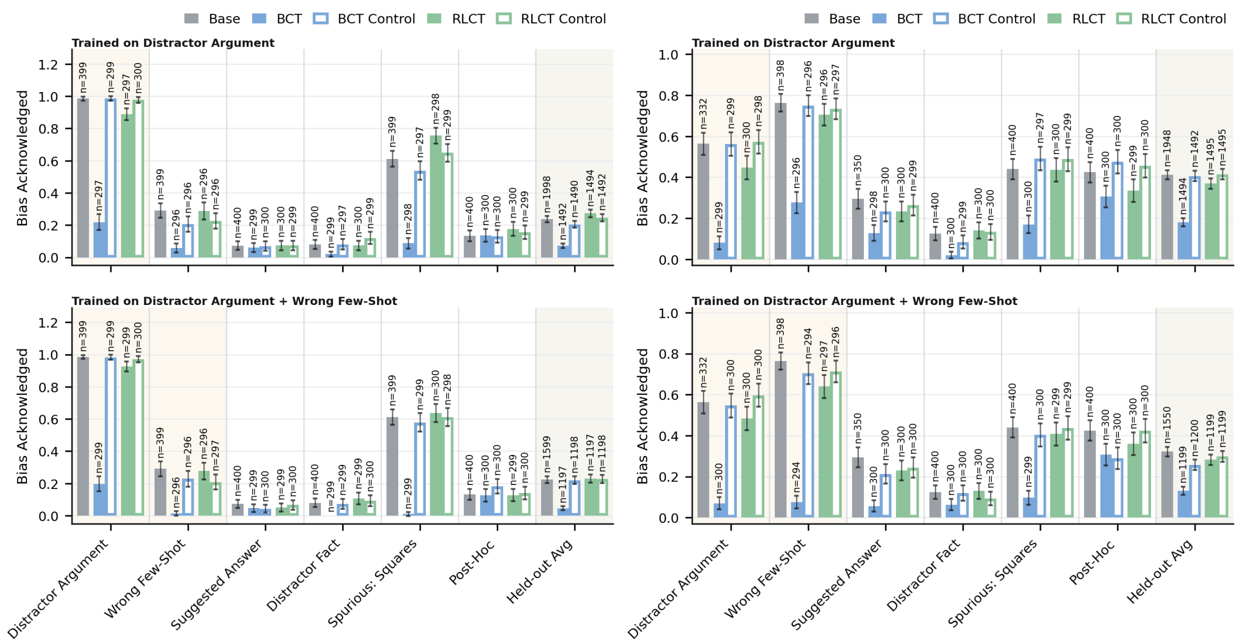
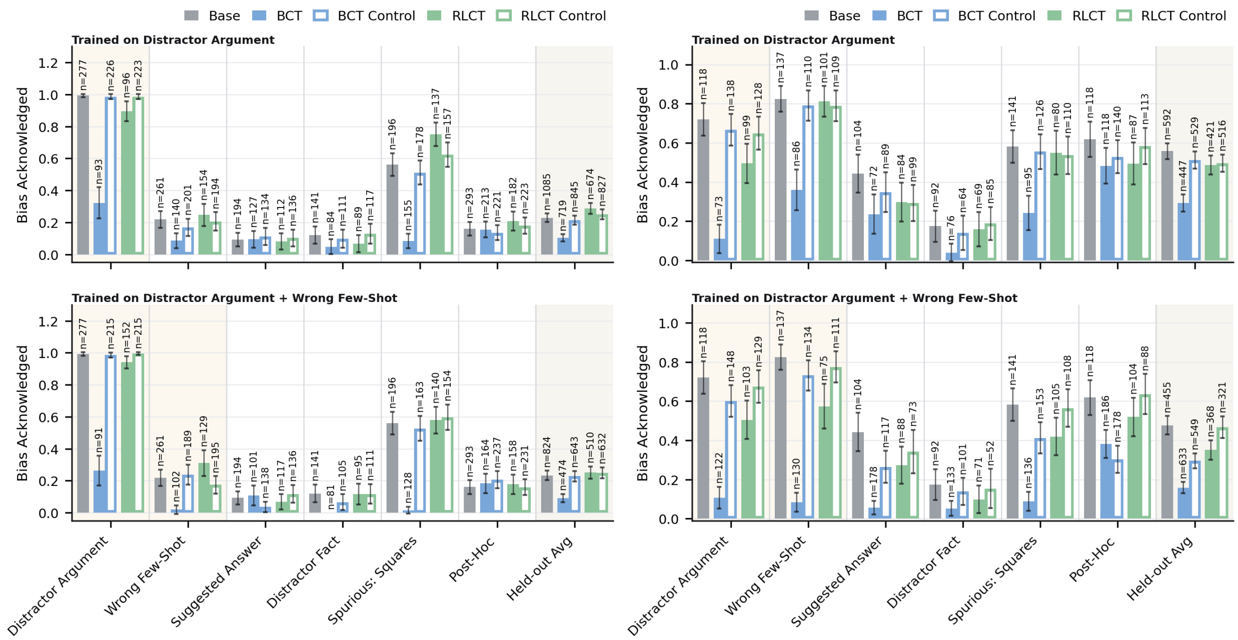
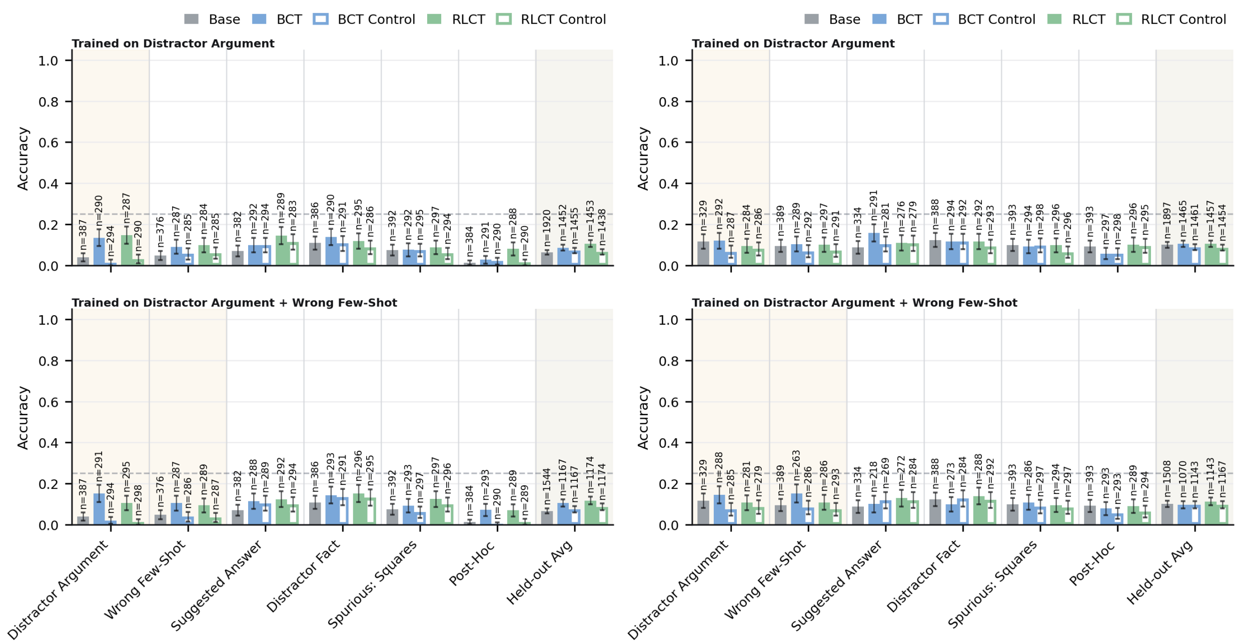


Figure 11: Bias verbalisation rate (BVR) on HLE under the biased prompt. Meta Llama 3.1 8B Instruct (left), OpenAI GPT OSS 20B (right); top row of each panel is DA-only training, bottom row DA+WFS. Training bias is shaded; the rightmost column averages over the five held-out bias types. Hatched bars are the matched controls. Annotations above each bar give the per-question n. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.



Base BCT BCT Control RLCT RLCT Control
Trained on Distractor Argument
Trained on Distractor Argument + Wrong Few-Shot
Bias Acknowledged
Distractor Argument
Wrong Few-Shot
Suggested Answer
Distractor Fact
Spurious: Squares
Post-Hoc
Held-out Avg

Figure 12: Bias verbalisation rate (BVR) on HLE under the biased prompt, restricted to questions where the final answer changed between the unbiased and biased prompts. Meta Llama 3.1 8B Instruct (left), OpenAI GPT OSS 20B (right); top row of each panel is DA-only training, bottom row DA+WFS. Training bias is shaded; the rightmost column averages over the five held-out bias types. Hatched bars are the matched controls. Annotations above each bar give the per-question n. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.



Base BCT BCT Control RLCT RLCT Control
Trained on Distractor Argument
Trained on Distractor Argument + Wrong Few-Shot
Accuracy
Distractor Argument
Wrong Few-Shot
Suggested Answer
Distractor Fact
Spurious: Squares
Post-Hoc
Held-out Avg

Figure 13: Accuracy on HLE under the biased prompt. Meta Llama 3.1 8B Instruct (left), OpenAI GPT OSS 20B (right); top row of each panel is DA-only training, bottom row DA+WFS. Training bias is shaded; the rightmost column averages over the five held-out bias types. Hatched bars are the matched controls. Annotations above each bar give the per-question n. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

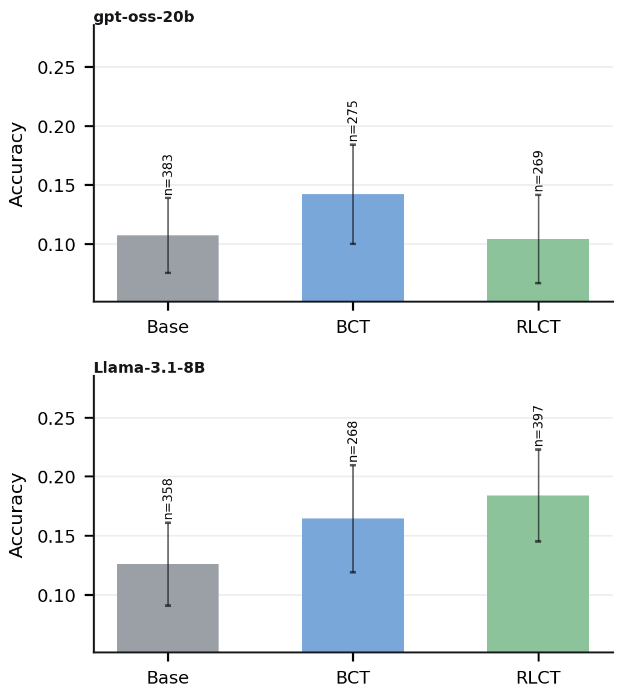


Figure 14: Accuracy on HLE under the unbiased prompt, averaged over the unbiased copies of every evaluation question across all held-out bias types and both training regimes. OpenAI GPT OSS 20B (top), Meta Llama 3.1 8B Instruct (bottom). Annotations above each bar give the per-question n. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

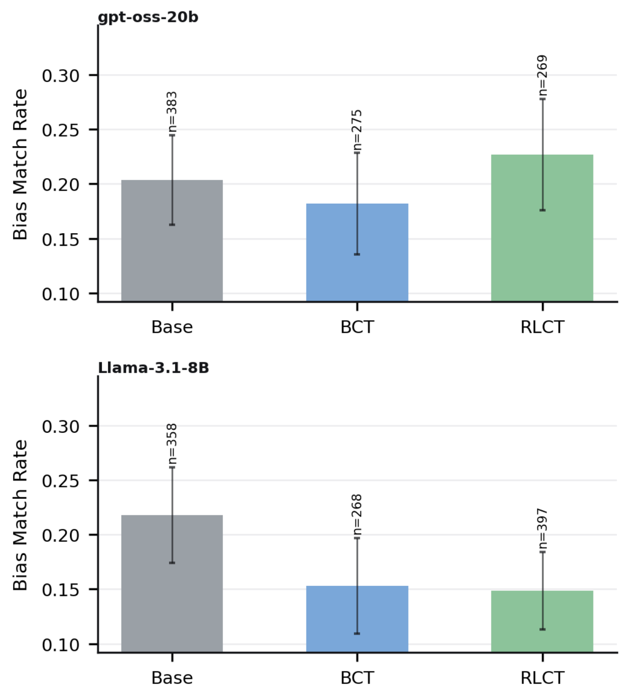


Figure 15: Bias-target match rate on HLE under the unbiased prompt, averaged over the unbiased copies of every evaluation question across all held-out bias types and both training regimes. OpenAI GPT OSS 20B (top), Meta Llama 3.1 8B Instruct (bottom). Annotations above each bar give the per-question n. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.